

AGENEWTECH

H1503BQ

Smart Module Specification_V1.7



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About the Document

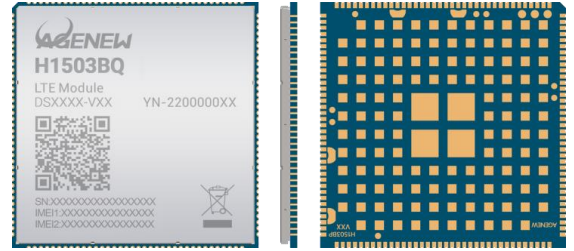
History

Version	Data	Author	Description
-	2024-05-21	Wu Danyi	Initial
1.0	2024-06-22	Wu Danyi	Controlled Copy
1.1	2024-08-23	Wu Danyi	Correct a mistake
1.2	2024-12-03	Yang Yanlian	Add frequency band allocation
1.3	2025-03-03	Yang Yanlian	Add frequency band allocation
1.4	2025-04-10	Yang Yanlian	Add frequency band allocation
1.5	2025-05-15	Wu Danyi	Integrate memory and radio frequency configurations
1.6	2025-07-04	Wu Danyi	Add text description for E2 frequency band
1.7	2026-03-23	Chen Yun	Change company name

H1503BQ

Multi-network Smart LTE module

Supports Wi-Fi & BT



H1503BQ is a multi-network Smart LTE Cat 4 module based on MediaTek's quad-core ARM Cortex-A53 processor launched by Agnew, equipped with Android 13 operating system. Its powerful performance, neon engine processing, 2D/3D graphics acceleration processing capabilities, fully meet the needs of customers for high-speed and multimedia functions in industrial and consumer applications.

H1503BQ supports full coverage of all network systems, WI-FI 802.11a/b/g/n/ac, and BT5.0 near-field wireless communication. In addition, the H1503BQ also has audio and video input and output interfaces and rich GPIO interfaces. It also supports a variety of audio and video codecs. The home screen supports HD+ 1280*800@60fps/720*1600@60fps at most. At the same time, H1503BQ supports GPS+GLONASS+BDS+Galileo positioning systems, making its positioning function fast and accurate.

H1503BQ is a patch module with a total of 275 pins, including 146 LCC pins and 129 LGA pins. The size is only 40.5mm × 40.5mm × 2.8mm, which can be embedded in various M2M product applications through pads. H1503BQ integrates rich interfaces (such as LCM, camera, touch screen, microphone, speaker, UART, USB, I2C, SPI, SDIO, and ADC interfaces), greatly expanding its application in the M2M field, making it widely used in smart cash registers, smart POS, tax control machines, security monitoring, vehicle equipment, and high-end information Collection equipment, smart robots, smart homes, smart hardware, industrial smart handheld devices, drones, high-end police or law enforcement equipment, smart intercom devices, smart wearables, vending machines, logistics cabinets and other equipment are used in different industries.



Main Advantages

- ✓ Comprehensive coverage of multi-network standard
- ✓ Support Wi-Fi 802.11a/b/g/n/ac and BT5.0 short-range wireless communication
- ✓ Support GPS/GLONASS/BDS/Galileo satellite positioning
- ✓ Support multiple audio and video codecs
- ✓ The main screen supports up to HD+ 1280×800@60fps/720×1600@60fps
- ✓ Support 1080P@30fps video recording and playback
- ✓ Possess audio and video input and output interfaces and rich GPIO interfaces



quad-core ARM Cortex-A53
64-bit processor



IEEE 802.11
a/b/g/n/ac



Support Android 13



GPS+GLONASS+BDS+Galileo



Powerful multimedia



BT5.0

LTE Cat4		H1503BQ			
Region/Operator	Eurasia/America				
Operating System	Android 13go/Android 13/Android 14go/Android 14				
Memory	1GB+8GB/1GB+16GB/2GB+32GB/3GB+32GB/4GB+64GB				
Module size	40.5mm(±0.1mm) × 40.5mm(±0.1mm) × 2.8mm(±0.2mm)				
Packaging	LCC + LGA				
Temperature Range					
Operation Temperature	Normal operating temperature: -20 °C ~ + 80°C				
	Storage temperature range: -40 °C ~ + 90 °C				
Frequency Band Information	-E1(P0)	-E2(P1)	-A1(P4)	-E3(P5)	-E4(P6)
LTE-FDD	B1/B3/B5/B7 /B8/B20/B28	B1/B2/B3/B4/B5 /B7/B8/B20/B28	B2/B4/B5/B7/B12/B13 /B17/B25/B26/B66/B71	B1/B3/B5/B7 /B8/ B20/B28	B1/B3/B5/B8
LTE-TDD	B34/B38/B39 /B40/B41(194M)	B38/B40/B41(120M)	B41(194M)	B38/B40/B41(120M)	B38/B40/B41(120M)
WCDMA	B1/B5/B8	B1/B2/B4/B5/B8	B2/B4/B5	B1/B5/B8	B1/B5/B8
GSM/EDGE	B2/B3/B5/B8	B2/B3/B5/B8	B2/B3/B5/B8	B3/B8	B3/B8
WLAN	2.4/5GHz, 802.11 a/b/g/n/ac				
Bluetooth	BT5.0				
GNSS	GPS+GLONASS+BDS+Galileo				
Authentication					
Operator Certification	Undetermined				
Compulsory Certification	CE				
Other Certifications	Undetermined				
Data Rate					
LTE-FDD (Mbps)	150Mbps (DL)/50Mbps (UL)				
LTE-TDD (Mbps)	150Mbps (DL)/50Mbps (UL)				
DC-HSPA+ (Mbps)	42Mbps (DL)/11.2Mbps (UL)				
WCDMA (Kbps)	384Kbps (DL)/384Kbps (UL)				
EDGE (Kbps)	296Kbps (DL)/236.8Kbps (UL)				
GPRS (Kbps)	107Kbps (DL)/85.6Kbps (UL)				
Outstanding characteristics					
Communication Mode	LTE: Release 12 UMTS: Release 9				
Bandwidth（MHz）	1.4/3/5/10/15/20				
Rx-Diversity	●				
DL supports MIMO 2 × 2	●				
SIM Card Detection	●				
Dual SIM Dual Standby	●				
USB Firmware Upgrade	●				
OTA Firmware Upgrade	●				

Remarks: 1. *: In progress
2. ●: Supports

LTE Cat4	H1503BQ
Electrical specification	
Operating Voltage	3.5~4.4 V, typical 3.8 V
Consumption (typical)	0.35mA@Shutdown, typical 2.8mA@sleep mode
Interface	
LCD	4-lane MIPI_DSI, HD+ 1280×800@60fps
Camera	Support 2 sets of camera, up to 21MP
TP	Capacitive touch screen
Audio	×3 Analog inputs: MIC1, MIC2, HEADMIC (×1 noise cancelling microphone) ×3 Analog outputs: differential linear (requires external audio amplifier), differential receiver and stereo earphone
USB	USB 2.0, Supports OTG, Charge
(U)SIM	×2, Supports 1.8/3V (U) SIM card, supports (U) SIM card hot plug detection, dual card dual standby
UART	×2
SD Card	×1
ADC	×2, 1 set general ADC interface, 1 set for battery ID
I2C	×6
SPI	×5
PWRKEY	Internal pull up to VBAT
GPIO	18 GPIO reserved
ANT	Main antenna, Diversity antenna, GNSS antenna, WIFI/BT antenna